

QUESTION No : 1

15/15 good.

1) $R1 = \{ (a, b) \mid a \text{ \& not\> } b \text{ have no letters in common} \}$

~~NOT~~ NOT REFLEXIVE, because we should have a loop such as (a, a) to derive reflexivity and since ~~no letters are common~~ in ~~a~~ a and b , the relation is not reflexive (5)

Relation is symmetric because if a and b have no letters in common then, b and a also do ~~not~~ have any letters in common. And since it's ~~not~~ symmetric, the relation is not anti-symmetric.

The relation is irreflexive because it is ~~not~~ reflexive on any pair.

If a, b have no letters in common and b, c have no letters in common, then a, c might or might not have the same letters, so relation can and cannot be transitive based on what we consider " c ".

2) $R2 = \{ (a, b) \mid a \text{ \& not\> } b \text{ are not the same length} \}$

The relation is symmetric because if (a, b) are ~~not~~ the same length, then (b, a) are also not of the same length. So, that makes this relation not anti-symmetric too.

The relation is NOT REFLEXIVE because (a, a) cannot exist because they cannot be of the same length. So, the relation also is irreflexive.

If (a, b) are not of same length \& not\> (b, c) are not of same length, then (a, c) ~~does not~~ necessarily have to be of same length. So, relation may or may not be transitive.

c) $R_3 = \{ (a, b) \mid a \text{ is longer than } b \}$

The relation is not symmetric because if $(a, b) \in R_3$ and a is longer than b , then $(b, a) \notin R_3$ because b has to be longer than a . So, the relation is anti-symmetric.

The relation is NOT REFLEXIVE because $(a, a) \notin R_3$ because a has to be longer than b and (a, a) are same.

So, the relation is irreflexive.

If $(a, b) \in R_3$ (a longer than b) and $(b, c) \in R_3$ (b longer than c), then $(a, c) \in R_3$ (a longer than c), so the relation is transitive.

QUESTION : 2

$$A = \{1, 2, 3\}$$

$$R = \{ (1, 1) (1, 3) (2, 1) (2, 2) (3, 1) \}$$

$$S = \{ (1, 2) (2, 1) (2, 2) (3, 3) \}$$

$$R = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \end{matrix}$$

$$S = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

$$R \cdot S = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix} \quad \text{by boolean product}$$

LIST:

$$R \cdot S = \{ (1, 2), (1, 3), (2, 1), (2, 2), (3, 2) \}$$

QUESTION No: 3

REFLEXIVE R . Show $R \subseteq R^2$

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R^2 is the composition of R with itself.

Such that pair (a,b) is in R^2 iff there exists a "c" in A , such that (a,c) and (c,b) are in R .

Now, R is reflexive so it has for example, (a,a) .

Suppose (a,b) is in R and (a,a) is in R as it is reflexive then by definition of R^2 , (a,a) in R implies that (a,b) is in R^2 , so basically $c=a$.

Therefore, for any element in R is also an element of R^2 , so $R \subseteq R^2$ is true.

QUESTION No. 4

5/5

SHOW EQUIVALENCE

The relation will be reflexive if for every aRa , $a-a$ results in an integer. Since, every $a-a$ will result in 0, and 0 is an integer, then R is reflexive.

The relation will be symmetric if for every aRb , then bRa . if $a-b$ is an integer, then $b-a$ is also an integer, which is true. So, relation R is symmetric.

The relation will be transitive if for every three elements in the set (a,b,c) , if aRb and bRc , then aRc , if $a-b$ and $b-c$ are integers then $a-c$ must also be an integer, which is true. So, R is transitive.

R is Reflexive, Symmetric, Transitive so it is equivalence relation.

QUESTION No: 5

INVERSE

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REFLEXIVE: if $(a, a) \in R$, then $(a, a) \in R^{-1}$ which makes it reflexive too. $\therefore$ inverse of pair

SYMMETRIC: if $(a, b) \in R$ and $(b, a) \in R$, then $(b, a) \in R^{-1}$ and $(a, b) \in R^{-1}$, which again is a symmetric relation. So, R^{-1} is symmetric.

TRANSITIVE: if $(a, b) \in R$ and $(b, c) \in R$ then $(a, c) \in R$, so $(b, a) \in R^{-1}$ and $(c, b) \in R^{-1}$ then $(c, a) \in R^{-1}$, which is again a transitive relation. So, R^{-1} is transitive.

QUESTION No: 6

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$$A = \{1, 2, 3\} \quad B = \{1, 2, 3, 4\}$$

$$R_1 = \{(1, 1), (2, 2), (3, 3)\} \quad R_2 = \{(1, 1), (1, 2), (1, 3), (1, 4)\}$$

$$\begin{aligned} R_1 \cup R_2 &= \{(1, 1), (2, 2), (3, 3)\} \cup \{(1, 1), (1, 2), (1, 3), (1, 4)\} \\ &= \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (3, 3)\} \end{aligned}$$

$$\begin{aligned} R_1 \cap R_2 &= \{(1, 1), (2, 2), (3, 3)\} \cap \{(1, 1), (1, 2), (1, 3), (1, 4)\} \\ &= \{(1, 1)\} \end{aligned}$$

$R_1 \oplus R_2$ (Symmetric Difference) but first,

$$\begin{aligned} R_1 - R_2 &= \{(1, 1), (2, 2), (3, 3)\} - \{(1, 1), (1, 2), (1, 3), (1, 4)\} \\ &= \{(2, 2), (3, 3)\} \end{aligned}$$

$$R_2 - R_1 = \{(1,1)(1,2)(1,3)(1,4)\} - \{(1,1)(2,2)(3,3)\}$$

$$= \{(1,2)(1,3)(1,4)\}$$

Now,

$$R_1 \oplus R_2 = (R_1 - R_2) \cup (R_2 - R_1)$$

{ From above found values }

$$= \{(2,2)(3,3)\} \cup \{(1,2)(1,3)(1,4)\}$$

$$= \{(1,2)(1,3)(1,4)(2,2)(3,3)\}$$

QUESTION No : 7

A = Set of Students

B = Set of Courses

$R_1 = \{(a,b) \mid a \text{ is a student who has taken course } b\}$

$R_2 = \{(a,b) \mid a \text{ is a student who requires course } b \text{ to graduate}\}$

$R_1 \cup R_2 =$ (includes all pairs either in R_1 or R_2)

Students who have taken a course or require a course to graduate.

$R_1 \cap R_2 =$ (includes only the pairs in both R_1 and R_2)

Students who have taken a course and require that course to graduate.

$R_1 \oplus R_2 =$ (includes pairs either in R_1 or R_2 but not in both)

Student has taken a course but does not need it to graduate or Student hasn't taken the course but needs it to graduate.

$R_1 - R_2 =$ (includes pairs in R_1 but not in R_2)

Student who have taken a course but does not need it to graduate

$\frac{5}{5}$ good.

$R_2 - R_1 =$ (includes pairs in R_2 but not in R_1)
Student who require a course to graduate but haven't taken that course.

QUESTION No : 8

$$R = \{(a, b) \mid a \equiv b \pmod{m}\} ; m > 1$$

REFLEXIVITY: for reflexive relation $(a, a) \in R$ and $a \equiv a \pmod{m}$ will always result in 0, no matter the value of m which is greater than 1 in our case. So, relation is reflexive.

SYMMETRY: for symmetric relation, if $(a, b) \in R$ then $(b, a) \in R$ and if $a \equiv b \pmod{m}$ is an integer, then $b \equiv a \pmod{m}$ must also be an integer. Since, the order of subtraction doesn't affect the result. So, it is also symmetric.

TRANSITIVITY: for transitive relation, if $(a, b) \in R$ and $(b, c) \in R$ then $(a, c) \in R$ and if $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$, then $a \equiv c \pmod{m}$, then by definition of congruent modulo m , the difference of two integers is also congruent modulo m . So, it is also transitive.

The Relation is equivalence relation as it is reflexive, symmetric, transitive, too.

QUESTION : 9

$$X = \{ \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\} \}$$

REFLEXIVITY : For R to be reflexive, for any subset A in X $A R A$, every subset A in X contains at least one element. So, the ~~least~~ least element of A is itself. Therefore, the relation is reflexive.

SYMMETRY : For R to be symmetric, if $A R B$, then $B R A$ for any subsets $A \cup B$ in X . If the least element of A equals the least element in B , then the least element of B equals the least element of A . Hence, the relation is symmetric.

TRANSITIVITY : For R to be transitive, if $A R B$ and $B R C$, then $A R C$ for any subsets $A, B \cup C$ in X , if least element of A equals least element of B and least element of B equals least element of C , then ultimately least element of A equals least element of C . Hence, the relation is transitive.

In conclusion, the relation is equivalence relation as it is reflexive, symmetric and transitive.

QUESTION : 10

$S = \{a, b, c, d, e\}$

aRb, bRc, dRe

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a) REFLEXIVE $\checkmark$ SYMMETRIC $\times$ TRANSITIVE $\times$

for every element x in S $(x, x) \in R$, so cRc is the only reflexive relation.

b) REFLEXIVE $\times$ SYMMETRIC $\checkmark$ TRANSITIVE $\times$

if $(x, y) \in R$ then $(y, x) \in R$, so cRb, bRa and eRd are the only symmetric relations according to the starting relations.

c) REFLEXIVE $\times$ SYMMETRIC $\times$ TRANSITIVE $\checkmark$

if $(x, y) \in R$ and $(y, z) \in R$, then $(x, z) \in R$, so in our case aRc is the only transitive relation.

d) for R to be equivalence relation, all points of relations must be true, which they are not in this case.
